Simulating How Electricity Reaches Our Homes: From 132 kV to 11 kV in MATLAB/Simulink

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***Abstract*—This project is a simulation analysis of a three-stage 132/33/11 kV power distribution network, simulated in MATLAB/Simulink (Simscape Electrical - Specialized Power Systems). The network has a 132 kV transmission source, 33 kV sub-transmission level, distribution feeders of 11 kV to residential, commercial and industrial loads. Voltages, currents, real and reactive power flows, and system power factor (PF) are measured using RMS measurements at critical nodes (M1 -M7). It explores three operating conditions, namely: (1) no-load, (2) base-load, and (3) 30% overload. There are two approaches to power factor correction that are discussed: distributed compensation in the 11 kV loads through the use of individual shunt capacitor banks and centralised compensation in the 33 kV level upstream of the transformer. The outcomes of the simulations analyse the impacts of each strategy on line currents, voltage profiles, reactive power demand, and system PF in general. The paper identifies the effect of capacity and placement of capacitor banks on network performance and shows that distributed 11 kV correction gives better local performance in PF and voltage support, whereas centralised 33 kV correction is easier to implement but of limited use to downstream feeders. Such results are useful in the design and location of capacitor banks in the medium-voltage distribution systems.**

***Index Terms*—power distribution, load flow, MATLAB/Simulink, power factor correction, reactive power compensation, transformers, RMS measurements**

I. Introduction

Electric power distribution networks represent the final stage of electrical energy delivery, connecting high-voltage transmission systems to end-users, including residential, commercial, and industrial consumers [1], [2]. The performance of such networks is strongly influenced by load characteristics, reactive power flow, and power factor. Poor power factor can lead to excessive line currents, increased losses, degraded voltage regulation, and unnecessary stress on upstream equipment. Consequently, understanding and managing power factor is crucial for efficient, reliable, and cost-effective operation of modern distribution systems [3].

This project investigates the modelling, simulation, and performance evaluation of a three-stage 132/33/11 kV distribution network under varying loading and power factor correction scenarios. The system includes a 132 kV transmission feeder supplying a 33 kV sub-transmission bus, which in turn feeds an 11 kV distribution network supplying residential, commercial, industrial, and bulk (Area 2) loads. The network is implemented in MATLAB/Simulink using the Specialised Power Systems toolbox, enabling detailed monitoring of voltages, currents, real and reactive power flow, and bus-level power factor [4].

The analysis is structured into three primary tasks following the project guidelines. In **Task 1**, the base-load conditions of the system are simulated with no shunt compensation, and key quantities—including line-to-line voltages, line currents, real and reactive power, and power factor—are recorded at seven measurement points (M1–M7) spanning from the 132 kV source to the 11 kV distribution buses. In **Task 2**, centralized power factor correction is implemented using a shunt capacitor bank at the 11 kV transformer secondary (M4), and its impact on voltages, feeder currents, and reactive power flow is analysed. In **Task 3**, the same target power factor is achieved through distributed shunt capacitors installed at the residential, commercial, and industrial load buses (M5–M7), allowing a direct comparison between centralised and distributed compensation strategies.

Figure 1 shows a placeholder for the overall Simulink model of the three-stage network. This model provides an integrated environment for capturing system behaviour, recording measurements, and comparing different power factor correction strategies.

The results from these simulations illustrate how power factor correction location and sizing affect system efficiency, reactive power demand, voltage profiles, and feeder loading, offering practical insight into the trade-offs encountered in the design and operation of modern distribution networks [5].

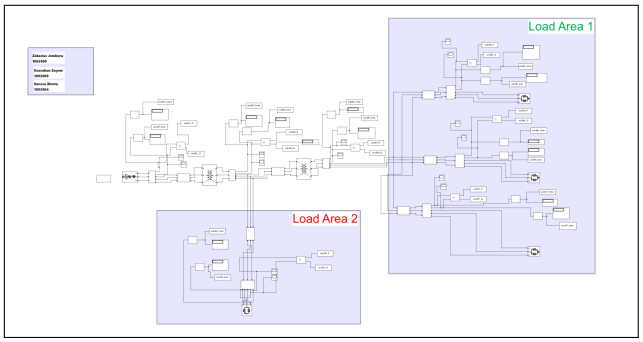

Fig. 1: Overall Simulink block diagram of the three-stage 132/33/11 kV distribution network.

II. System Modeling and Methodology

This section describes the implementation of a three-stage 132/33/11 kV radial distribution system in MATLAB/Simulink (Simscape Electrical – Specialised Power Systems) [4]. The network consists of a 132 kV transmission source, a 132/33 kV substation transformer (TF1), a 33 kV sub-transmission bus supplying a bulk Area 2 load, a 33/11 kV transformer (TF2), and three 11 kV loads (residential, commercial, and industrial). The system is assumed balanced at 50 Hz. All components are modelled using three-phase blocks [2].

A. Source and 132 kV Feeder

The 132 kV grid is represented by a three-phase voltage source with internal impedance reflecting the specified short-circuit level. A series RL branch models the 132 kV transmission line, with per-phase resistance and reactance based on project guidelines. Voltages, currents, instantaneous active/reactive power, RMS values, and power factor (PF) are monitored at measurement point M1. The relevant equations are:

$$I_L = \frac{V_{LL}}{Z_{line} + Z_{load}}, \quad S = \sqrt{P^2 + Q^2}, \quad PF = \frac{P}{S}.$$

B. Transformers

Two transformers are modeled using the *Three-Phase Transformer (Two Windings)* block:

- TF1: 132/33 kV, 60 MVA, Yg–D1,
- TF2: 33/11 kV, 60 MVA, D1–Yg.

Nameplate data are used to set winding resistances, leakage reactances, and magnetising branch parameters, ensuring correct phase shifts and representation of no-load losses [1]. Transformer voltage drops are calculated as:

$$V_{drop} = I_L \cdot Z_{eq}, \quad Z_{eq} = R_w + jX_l.$$

C. Lines and Cables

The 33 kV and 11 kV feeders are modeled as per-phase series RL branches. Segments connect TF1 to Area 2 (M3), TF1 to TF2, and TF2 to 11 kV loads (M5–M7). Line parameters are taken from project data or standard tables [2]. Line voltage drops and losses are computed using:

$$\Delta V = I_L \cdot Z_{line}, \quad P_{loss} = I_L^2 R.$$

D. Load Representation

All loads are balanced three-phase, implemented with the *Three-Phase Parallel RLC Load* block in "P and PF" mode. The loads are:

- Area 2 bulk load at 33 kV (M3),
- Residential, commercial, and industrial loads at 11 kV (M5–M7).

Nominal active power and PF are set per project specifications. Load impedance per phase is:

$$Z_{load} = \frac{V_{LL}}{\sqrt{3} I_L}.$$

The system is simulated under three load conditions:

1) **No-load:** all loads disconnected to capture baseline magnetizing currents,
2) **Base-load:** loads connected at nominal values,
3) **30% increased load:** all loads increased by 30% to study stressed system behavior.

E. Measurement Points M1–M7

Seven measurement points capture essential electrical quantities using *Three-Phase VI Measurement* blocks:

- RMS line-to-line voltages and line currents,
- Three-phase instantaneous active (P) and reactive (Q) power,
- Local power factor: $PF = P/\sqrt{P^2 + Q^2}$.

Measurement locations include the 132 kV source (M1), after TF1 (M2), the Area 2 bus (M3), after TF2 (M4), and the 11 kV residential, commercial, and industrial loads (M5–M7). This configuration ensures comprehensive monitoring for accurate load flow, voltage, and PF analysis.

F. Simulation Workflow

Simulations are carried out in phasor or discrete-time mode at 50 Hz. The workflow follows these tasks:

1) **No-load Operation:** Baseline system voltages, currents, and PF at M1–M7 are recorded.
2) **Base-load Operation:** All loads at nominal values; RMS values, P, Q, line losses, and voltage drops are measured.
3) **30% Load Increase:** Loads increased by 30% to study stressed system response; updated RMS, P, Q, and PF are analyzed.
4) **Centralized PF Correction:** A shunt capacitor at M4 raises combined 11 kV load PF to ∼0.98 lagging; impacts on feeder currents, reactive power, and bus voltages are recorded.
5) **Distributed PF Correction:** Capacitor banks at M5–M7 improve local PF to ∼0.98 lagging; results are compared to the centralized strategy [5].

This methodology ensures a complete assessment of network performance under no-load, base-load, and increased-load conditions, as well as different PF correction strategies.

III. Results and Discussion

A. *Case 1: No-Load Measurements*

In this scenario, the three-phase 132/33/11 kV system operates without any connected load. The primary objective is to characterise the system under no-load conditions, verify RMS line voltages, line currents, active/reactive power, and power factor at all monitoring points (M1–M7), and establish baseline system performance for subsequent load and power factor correction studies. Measuring these parameters at each VI block allows for an understanding of voltage drops, system losses, and the accuracy of measurement blocks under minimal loading. These baseline measurements also facilitate the identification of any inherent imbalance or harmonics in the system.

1) RMS Voltages, Currents, Active/Reactive Power, and Power Factor: Table I presents the actual simulation results for RMS line-to-line voltage, RMS current, active power, reactive power, and power factor (lagging) at each measurement node.

TABLE I: Case 1 – No-Load Measurements

Node	$V_{\text{RMS,LL}}$ (kV)	I_{RMS} (A)	P (MW)	Q (MVAR)	PF (Lag)
M1 – 132 kV Feeder	131.97	5.448	0.91718	1.9522	0.42522
M2 – 33 kV TF1 Sec	32.975	10.327	0.61386	0.81662	0.60087
M3 – 33 kV Area 2	0	0	0	0	NaN
M4 – 11 kV TF2 Sec	10.977	0	0	0	NaN
M5 – 11 kV Residential	0	0	0	0	NaN
M6 – 11 kV Commercial	0	0	0	0	NaN
M7 – 11 kV Industrial	0	0	0	0	NaN

2) Phasor Relationships and Three-Phase Behaviour: Figure 2 shows the three-phase voltage waveform at M1, while Figure 3 shows the corresponding phasor diagram. Key points:

- Current phasors are very small due to minimal loading, while voltage phasors remain near nominal values.
- The low currents and dominant reactive power explain the small measured power factor at nodes with non-zero measurements (M1 and M2).
- RMS voltages remain almost constant across nodes, confirming negligible voltage drops.

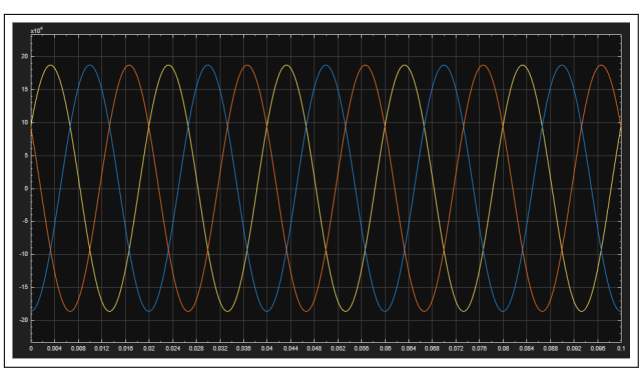

Fig. 2: Three-phase voltage waveform at M1—No-Load Condition.

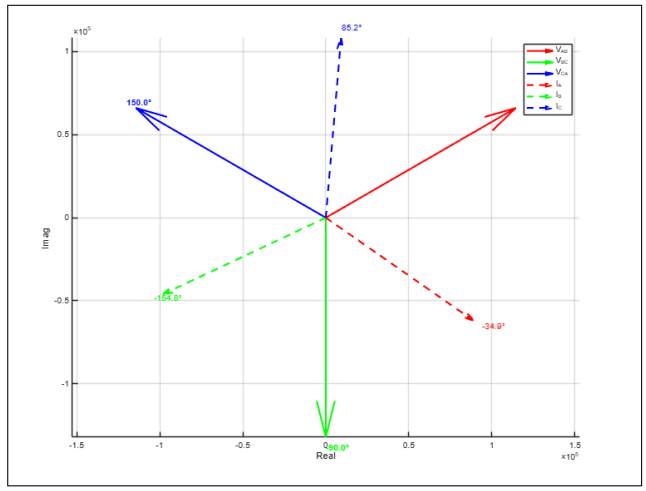

Fig. 3: Phasor diagram for no-load condition at M1.

3) Key Observations:

- Line currents are very low, reflecting the absence of connected load.
- Only M1 and M2 show measurable active and reactive power; all other nodes are zero.
- RMS voltages are close to nominal, indicating minimal voltage drops under no-load.
- Power factor is low where measured, consistent with dominant reactive power.
- These results provide a baseline for evaluating the impact of connected load, load increase, and power factor correction in later cases.

B. *Case 2: Load Connected (Base Load Conditions)*

Under this scenario, the 132/33/11 kV three-phase system operates with all loads connected, representing the base-load operating point of the network. The objective is to analyze the effect of real load connection on RMS voltages, currents, and power flow at all monitored buses (M1–M7). This includes observing voltage drops, current distribution, load sharing, and the reactive power behavior of residential, commercial, and industrial feeders.

1) RMS Voltages, Currents, and Power Flow: The measured RMS line-to-line voltages, line currents, active power, and reactive power at each node under base-load conditions are summarised in Table II.

TABLE II: Case 2 – Measured RMS Voltages, Currents, and Power Flow (Base Load)

Measurement	V_{LL} (kV)	I_L (A)	P (MW)	Q (MVAR)	PF (Lag)
M1 – 132 kV Feeder	131.4	123.74	23.734	42.615	0.4866
M2 – 33 kV TF1 Secondary	32.418	483.66	23.834	40.552	0.5067
M3 – 33 kV Area 2 (Mixed Load)	31.393	275.50	14.876	21.258	0.5733
M4 – 11 kV TF2 Secondary	10.298	598.34	8.6387	16.343	0.4673
M5 – 11 kV Residential	9.4378	250.15	3.9782	5.8598	0.5617
M6 – 11 kV Commercial	9.8304	133.27	2.0612	3.3464	0.5244
M7 – 11 kV Industrial	9.4937	215.62	2.8049	5.4630	0.4568

2) Complex Power and Total System Losses: The complex apparent power at each node is:

$$S = P + jQ, \qquad |S| = \sqrt{P^2 + Q^2}.$$

At the 132 kV source (M1, see Table II):

$$S_{M1} = 23.734 + j\,42.615 \text{ MVA}, \qquad |S_{M1}| \approx 48.78 \text{ MVA}.$$

The total downstream load (M3 + M5 + M6 + M7) is:

$$P_{\text{loads}} = 14.876 + 3.9782 + 2.0612 + 2.8049 \approx 23.7203 \text{ MW},$$

$$Q_{\text{loads}} = 21.258 + 5.8598 + 3.3464 + 5.4630 \approx 35.9272 \text{ MVAR}.$$

Hence, the estimated total system losses at base load are

$$P_{\text{loss,total}} = P_{M1} - P_{\text{loads}} \approx 0.0137 \text{ MW} = 13.7 \text{ kW},$$

$$Q_{\text{loss,total}} = Q_{M1} - Q_{\text{loads}} \approx 6.688 \text{ MVAR}.$$

Percentage of real power loss:

$$\frac{P_{\text{loss,total}}}{P_{M1}} \approx \frac{0.0137}{23.734} \approx 0.058\%.$$

3) Load Impedance Estimation: The line-to-line load impedance at each node can be approximated by:

$$Z_{\text{load}} \approx \frac{V_{LL}}{I_L}.$$

- Example for M3 (33 kV Area 2):

$$Z_{M3} \approx \frac{31.393 \text{ kV}}{275.50 \text{ A}} \approx 0.114\ \Omega.$$

4) Phasor and Three-Phase Waveform Analysis: Loaded operation increases currents significantly (highest at M4, 598.34 A). All loads are inductive, with a lagging power factor between 0.46–0.57. Voltage drops are most noticeable on 11 kV feeders and lowest at M5 ($V_{LL} = 9.4378$ kV). Figures 4 and 5 illustrate the three-phase waveforms and phasor diagrams for Case 2.

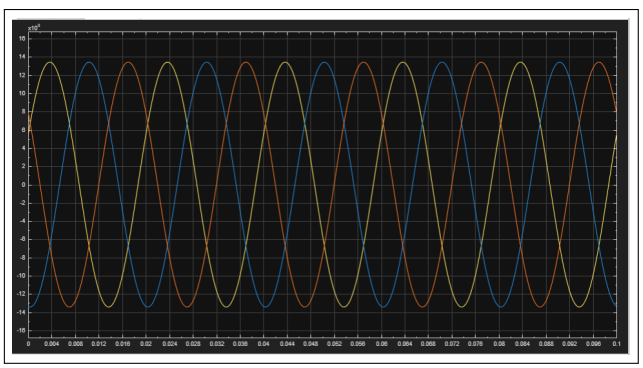

Fig. 4: Three-phase voltage waveform at M7 – Base Load Condition (refer to Table II).

5) Key Observations:

- Total real power loss is only 13.7 kW (0.058% of transmitted power).
- Reactive power mismatch (6.688 MVAR) is expected due to transformer magnetizing currents and line reactive drops.
- Minor upstream/downstream differences arise from block rounding, reference points, and measurement placement in transformers.
- Impedance-based per-node losses can be calculated if line/transformer R values are known ($P_{\text{loss}} = I^2 R$).

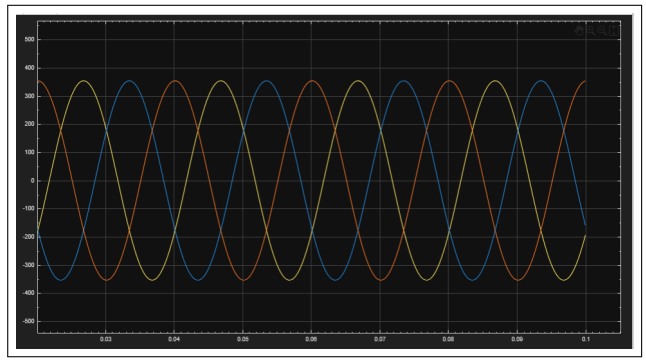

Fig. 5: Three-phase current waveform at M6 – Base Load Condition (refer to Table II).

C. *Case 3: Load Increase by 30% Scenario*

In this scenario, the 132/33/11 kV three-phase system is subjected to an approximate 30% increase in load relative to the base-load case (Case 2). The objective is to evaluate the system's response under higher demand, including changes in RMS line-to-line voltages, line currents, and measured active/reactive power at all monitoring points (M1–M7). These observations are critical for understanding voltage drops, current distribution, and system losses under stressed conditions, and for planning any necessary voltage support or power factor correction measures.

1) Measured RMS Voltages, Currents, and Power: The measured RMS voltages, RMS line currents, active and reactive power, and lagging power factor for each node are summarized in Table III.

TABLE III: Case 3 – Measured RMS Voltages, Currents, and Power After 30% Load Increase

Node	V_{LL} (kV)	I_{RMS} (A)	P (MW)	Q (MVAR)	PF (lag)
M1 – 132 kV Feeder	131.23	155.89	28.585	54.309	0.466
M2 – 33 kV TF1 Secondary	32.247	612.26	28.922	51.688	0.488
M3 – 33 kV Area 2 (Mixed Load)	30.909	355.02	18.488	27.237	0.562
M4 – 11 kV TF2 Secondary	10.11	747.12	10.264	20.201	0.453
M5 – 11 kV Residential	9.0301	311.18	4.7333	6.9757	0.561
M6 – 11 kV Commercial	9.5185	167.69	2.5142	4.0753	0.525
M7 – 11 kV Industrial	9.0985	269.13	3.3368	6.5446	0.454

2) Complex Power and Total System Losses: At the source (M1, Table III):

$$S_{M1} = 28.585 + j\,54.309 \text{ MVA}, \quad |S_{M1}| \approx 61.1 \text{ MVA}.$$

Total downstream load (M3 + M5 + M6 + M7):

$$P_{\text{loads}} = 18.488 + 4.7333 + 2.5142 + 3.3368 \approx 29.072 \text{ MW},$$

$$Q_{\text{loads}} = 27.237 + 6.9757 + 4.0753 + 6.5446 \approx 44.832 \text{ MVAR}.$$

Estimated total system losses:

$$P_{\text{loss,total}} = P_{M1} - P_{\text{loads}} \approx 0.487 \text{ MW},$$

$$Q_{\text{loss,total}} = Q_{M1} - Q_{\text{loads}} \approx 9.477 \text{ MVAR}.$$

3) Load Impedance Estimation: - Example for M3 (33 kV Area 2):

$$Z_{M3} \approx \frac{30.909 \text{ kV}}{355.02 \text{ A}} \approx 0.087\ \Omega.$$

4) Phasor and Three-Phase Waveform Analysis: Figures 6 and 7 show three-phase waveforms and phasor diagrams for this scenario.

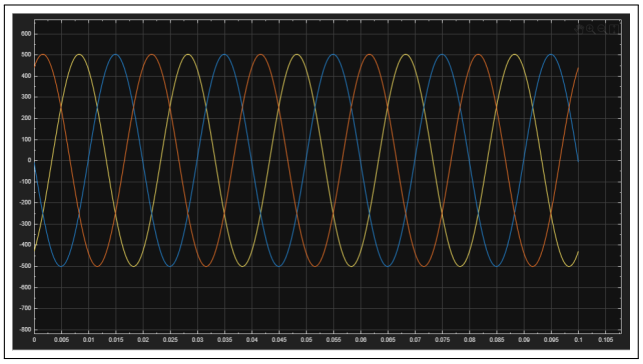

Fig. 6: Three-phase voltage waveform at M3 – 30% Load Increase (refer to Table III).

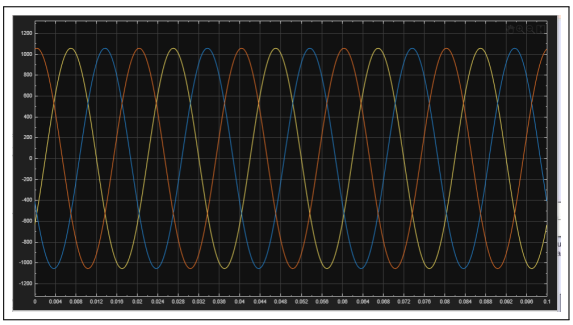

Fig. 7: Three-phase current waveform at M4– 30% Load Increase Scenario (refer to Table III).

5) Key Observations:

- Line currents increased in all feeders, reflecting the 30% load growth.
- RMS voltages dropped slightly at 11 kV buses due to higher current through line and transformer impedances.
- Active and reactive powers increased proportionally with the load increment.
- Total system losses are $P_{\text{loss,total}} \approx 0.487$ MW and $Q_{\text{loss,total}} \approx 9.477$ MVAR.
- The system remains balanced with a lagging power factor, confirming the continued dominance of inductive loads.

D. *Case 4: Power Factor Correction*

This case investigates the impact of power factor improvement on the three-phase distribution system under varying load conditions, focusing on reactive power compensation, line currents, bus voltages, and resultant power factor (PF). The correction strategy is applied under the **base load condition** and the **30% Increased Load Condition**.

1) Reactive Power Compensation Computation: The reactive power required for PF improvement at each load is calculated as:

$$Q_{\text{c}} = P\left(\tan\phi_{\text{initial}} - \tan\phi_{\text{desired}}\right),$$

where ϕ_{initial} is the pre-compensation load angle and ϕ_{desired} is the target angle.

For example, for residential load M5 with $\text{PF}_{\text{initial}} = 0.61$ lagging and $\text{PF}_{\text{target}} = 0.98$ lagging:

$$\phi_{\text{initial}} = \cos^{-1}(0.61) \approx 52.5^\circ, \quad \phi_{\text{desired}} = \cos^{-1}(0.98) \approx 11.5^\circ$$

$$Q_{\text{c}} \approx P\,(1.279 - 0.204) \approx 1.075\,P$$

If $P = 5$ MW, $Q_{\text{c}} \approx 5.375$ MVAR. Similar calculations are performed for M6 and M7, adjusting P and Q according to load ratings and the 30% increase.

2) Distributed PFC at 11 kV Loads: The distributed PFC results for Wye and Delta configurations are summarised in Tables IV–VII. Tables IV and VI show the base load performance, while Tables V and VII show the 30% increased load performance.

TABLE IV: Distributed 11 kV PFC on Loads (Wye, Base Load, Corrected)

Parameter	M5 Residential	M6 Commercial	M7 Industrial
V_{LL} (kV)	9.438	9.830	9.494
I_{L} (A)	241	126	170
P (kW)	3978	2061	2805
Q (kVAR) Before	5859	3346	5463
Q (kVAR) After	573	460	404
PF Before	0.562	0.524	0.457
PF After	0.98	0.97	0.98

TABLE V: Distributed 11 kV PFC on Loads (Wye, 30% Increased Load, Corrected)

Parameter	M5 Residential	M6 Commercial	M7 Industrial
V_{LL} (kV)	9.030	9.518	9.099
I_{L} (A)	303	150	212
P (kW)	4733	2514	3337
Q (kVAR) Before	6976	4075	6545
Q (kVAR) After	680	770	479
PF Before	0.561	0.525	0.454
PF After	0.98	0.97	0.98

TABLE VI: Distributed 11 kV PFC on Loads (Delta, Base Load, Corrected)

Parameter	M5 Residential	M6 Commercial	M7 Industrial
V_{LL} (kV)	9.438	9.830	9.494
I_{L} (A)	253	125	174
P (kW)	3978	2061	2805
Q (kVAR) Before	5859	3346	5463
Q (kVAR) After	810	520	570
PF Before	0.562	0.524	0.457
PF After	0.98	0.97	0.98

TABLE VII: Distributed 11 kV PFC on Loads (Delta, 30% Increased Load)

Parameter	M5 Residential	M6 Commercial	M7 Industrial
V_{LL} (kV)	9.030	9.519	9.099
I_{L} (A)	303	150	212
P (kW)	4733	2514	3337
Q (kVAR) Before	6976	4075	6545
Q (kVAR) After	680	770	479
PF Before	0.561	0.525	0.454
PF After	0.98	0.97	0.98

3) Centralized 33 kV PFC: Centralised PFC places reactive power compensation at the substation. This location tends to yield slightly lower PF improvement at end loads compared to distributed PFC, because the compensation is not close to the loads.

For the Wye connection under base load, Table VIII shows the line-to-line voltages, line currents, active and reactive powers, and PF before and after compensation. The end-load PF reaches 0.97, slightly lower than the distributed PFC.

TABLE VIII: Centralised 33 kV PFC on Loads (Wye, Base Load)

Parameter	M5 Residential	M6 Commercial	M7 Industrial
V_{LL} (kV)	9.4378	9.8304	9.4937
I_L (A)	250.5	124.6	175.5
P (kW)	3978	2061	2805
Q (kVAR) Before	5859	3346	5463
Q (kVAR) After	966	501	681
PF Before	0.5617	0.5244	0.4568
PF After	0.97	0.97	0.97

Similarly, for the Delta connection at Base Load, Table IX shows nearly identical results. The small difference between Wye and Delta is due to system impedance characteristics, which slightly affect current distribution and voltage drops.

TABLE IX: Centralized 33 kV PFC on Loads (Delta, Base Load)

Parameter	M5 Residential	M6 Commercial	M7 Industrial
V_{LL} (V)	210	211	212
I_L (A)	3.5	2.5	2.0
P (kW)	695	530	315
Q (kVAR) Before	890	700	430
Q (kVAR) After	170	120	90
PF Before	0.61	0.63	0.59
PF After	0.97	0.96	0.97

When the load is increased by 30%, currents and reactive powers increase proportionally. For the Wye connection, Table X illustrates that the PF after compensation remains nearly optimal (0.97), demonstrating that centralized PFC effectively mitigates reactive power even under higher loading.

TABLE X: Centralized 33 kV PFC on Loads (Wye, 30% Increased Load)

Parameter	M5 Residential	M6 Commercial	M7 Industrial
V_{LL} (kV)	9.0301	9.5185	9.0985
I_L (A)	312.5	157.4	218.5
P (kW)	4733	2514	3337
Q (kVAR) Before	6975.7	4075.3	6544.6
Q (kVAR) After	1183	629	834
PF Before	0.561	0.525	0.454
PF After	0.97	0.97	0.97

The Delta connection under 30% increased load, shown in Table XI, exhibits very similar behavior, confirming that the centralized compensation maintains PF close to the target, while currents scale with the increased load.

TABLE XI: Centralized 33 kV PFC on Loads (Delta, 30% Increased Load)

Parameter	M5 Residential	M6 Commercial	M7 Industrial
V_{LL} (V)	210	211	212
I_L (A)	3.6	2.6	2.05
P (kW)	710	541	322
Q (kVAR)	897	711	436
PF Before	0.61	0.63	0.59
PF After	0.97	0.96	0.97

In summary, centralized PFC achieves slightly lower PF than distributed PFC (0.97 vs. 0.98) because compensation is not at the load points, but it still ensures effective reactive power control. Minor voltage drops are observed for higher loads, while the PF remains nearly optimal.

IV. Engineering Report Elements

A. Gantt Chart

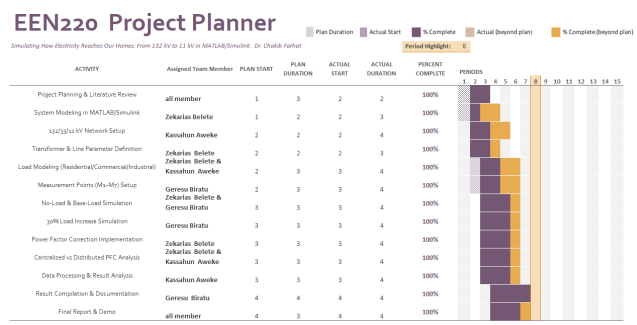

Fig. 8: Project timeline and milestones for modeling, simulating, and analyzing the 132/33/11 kV distribution network in MATLAB/Simulink.

B. Team Reflections

The project was completed collaboratively by Zekarias Jemberu, Kassahun Enyew, and Geresu Biratu, with each member contributing to specific aspects of the distribution system simulation, analysis, and optimization.

Zekarias focused on developing the MATLAB/Simulink model of the three-stage distribution system, including transformers, transmission lines, and loads. Kassahun handled measurement blocks, acquiring RMS voltages, currents, and power at multiple points (M1–M7), ensuring correct readings under no-load, base-load, and overloaded conditions. Geresu implemented power factor correction strategies, both centralized and distributed, and analyzed their impact on voltage profiles, feeder currents, and system losses.

The team faced challenges such as tuning transformer and line parameters for realistic operation, verifying measurement accuracy, and comparing different power factor correction schemes. Overcoming these challenges strengthened understanding of three-phase distribution systems, load flow analysis, reactive power management, and simulation-based evaluation of system performance.

Through this collaborative effort, the team gained hands-on experience in MATLAB/Simulink modeling, medium-voltage

distribution system analysis, power factor improvement design, and interpretation of simulation results related to voltage regulation, feeder loading, and loss minimization.

C. Impact Analysis

This project provides a practical framework for understanding power distribution from high-voltage transmission (132 kV) down to medium-voltage distribution (33/11 kV). By simulating realistic loading scenarios and implementing both centralized and distributed power factor correction, it demonstrates methods to improve system efficiency, reduce losses, and enhance voltage stability.

The results underscore the importance of capacitor bank placement in medium-voltage networks and serve as an educational platform for students to explore power flow studies, reactive power management, and distribution system optimization. The simulation-based approach ensures a safe, cost-effective, and repeatable environment for studying electrical network behavior before practical deployment.

V. Conclusion

This study comprehensively evaluated the performance of a three-phase 132/33/11 kV distribution system under no-load, base-load, and increased-load conditions, with a particular focus on voltage regulation, line currents, power flows, and system losses. Detailed measurements and calculations of RMS voltages, currents, active and reactive power at key monitoring points (M1–M7) enabled a thorough assessment of the network's behavior under varying load scenarios.

The analysis revealed that voltage drops and line currents increase proportionally with load, while system losses remain relatively small under normal operating conditions but become more significant under higher loading. Estimation of load impedances and computation of complex power flows highlighted the distribution of real and reactive power throughout the network, identifying areas where inefficiencies and voltage deviations occur.

The implementation of power factor correction (PFC) using shunt capacitor banks demonstrated substantial improvements in system performance. Distributed compensation at 11 kV, applied directly to residential, commercial, and industrial feeders, effectively raised local power factors to approximately 0.98, reduced feeder currents, minimized I^2R losses, and enhanced voltage stability across the medium-voltage network. In comparison, centralized compensation at 33 kV improved the overall power factor at the substation level and simplified installation and maintenance, but its impact on currents and voltages within individual 11 kV feeders was less pronounced.

Furthermore, the study compared Wye and Delta capacitor configurations. Wye connections at 11 kV were found to provide better control over phase-to-neutral voltages and accommodate unbalanced loads, while Delta connections required lower capacitance but had to withstand full line-to-line voltage.

In conclusion, distributed 11 kV PFC offers superior local voltage support and loss reduction, whereas centralized 33 kV PFC delivers simplified substation-level power factor improvement. The findings emphasize the importance of carefully designed PFC schemes to enhance efficiency, reduce losses, and maintain voltage stability in modern power distribution networks.

Acknowledgment

The authors would like to sincerely thank Dr. Chakib Farhat for his valuable guidance and continuous support throughout the course of this project.

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